

DATABASE SECURITY

A Presentation by : Amir Khanzada

Roll No: 2k11/SWE/24



OUTLINE

- ❑ Overview To Database Security.
- ❑ What is Database Security
- ❑ Why need of database security.
- ❑ Concepts of Database Security.
- ❑ Security Problems
- ❑ Security Controls

OVERVIEW

In today's world, we need everything secured whether it is your mobile phone , computer , vehicle or almost anything.



Mobile



Computer



Vehicles

What is database security?

Database:

It is a collection of information stored in a computer.

Security:

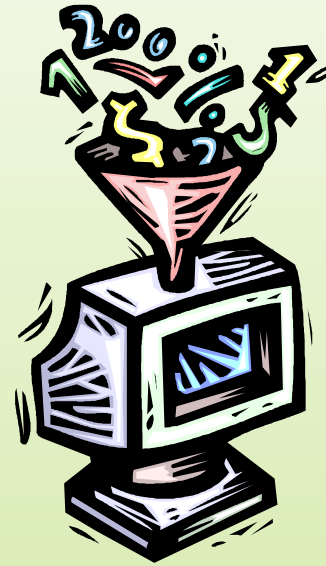
It is being free from danger.

Database Security:

It is the mechanisms that protect the database against intentional or accidental threats.

Definition of Database Security

Database Security is defined as the process by which “Confidentiality, Integrity and Availability” of the database can be protected



Why need of database security?

If there is no security to database what happens???



Data will be easily corrupted

It is important *to restrict access to the database from authorized users* to protect sensitive data.

Concepts of Database Security

There are 3 main aspects

1. Secrecy or Confidentiality
2. Integrity
3. Availability

SECRECY / CONFIDENTIALITY

- It is protecting the database from unauthorized users.
- Ensures that users are allowed to do the things they are trying to do.
- Encryption is a technique or a process by which the data is encoded in such a way that only that authorized users are able to read the data.

INTEGRITY

- Protecting the database from authorized users.
- Ensures that what users are trying to do is correct.
- For examples,
 - An employee should be able to modify his or her own information.

AVAILABILITY

- Database must have not unplanned downtime.
- To ensure this ,following steps should be taken
- ***Restrict*** the amount of **the storage space** given to each user in the database.
- ***Limit*** the number of ***concurrent sessions*** made available to each database user.
- ***Back up*** the data ***at periodic intervals*** to ensure data recovery in case of application users.



SECURITY PROBLEMS



- *Any circumstance or event with the potential to adversely impact an IS through unauthorized access, destruction, disclosure, modification of data, and/or denial of service.*

There are two kinds of threat.

- **Non-fraudulent Threat**

- **fraudulent Threat**

1. Non-fraudulent Threat

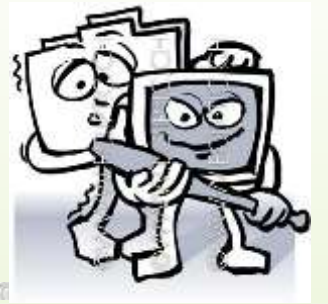
- Natural or accidental disasters.
- Errors or bugs in hardware or software.
- Human errors.

2. fraudulent Threat

- Authorized users
 - Those who abuse their privileges and authority.
- Hostile agents
 - Those improper users (outsider or insiders).
 - who attack the software and/or hardware system, or read or write data in a database.

DATABASE PROTECTION REQUIREMENTS

1. Protection from Improper Access
2. Protection from Inference
3. Integrity of the Database
4. User Authentication
5. Multilevel Protection
6. Confinement
7. Management and Protection of Sensitive Data



SECURITY CONTROLS



METHODS OF SECURING THE DATABASE

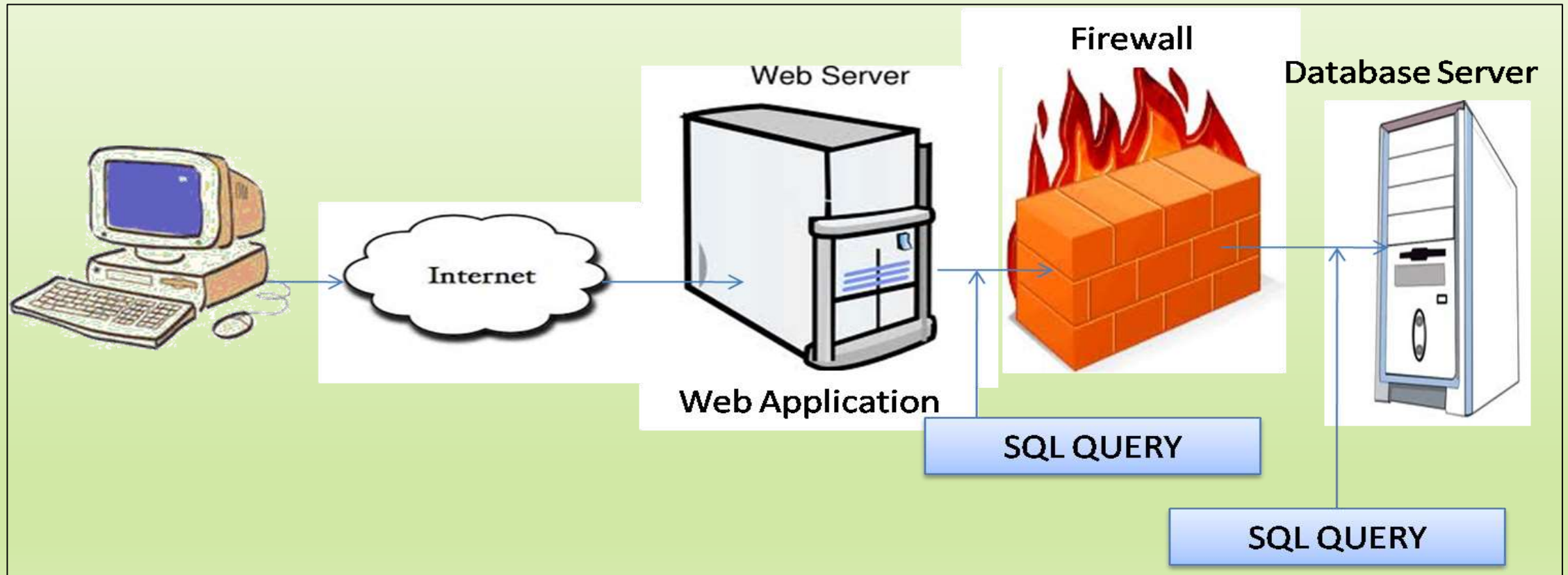
- *Authorization* - privileges, views.
- *Encryption* - public key / private key, secure sockets.
- *Authentication* – passwords.
- *Logical* - firewalls, net proxies.

SECURITY OF THE DATABASE THROUGH FIREWALLS

A FIREWALL is dedicated *software* on another computer *which inspects network traffic* passing through it and denies (*or*) permits passage based on set of rules. Basically it is a piece of software that monitors all traffic that goes from your system to another via the Internet or network and Vice Versa

Database Firewalls are a type of Web Application Firewalls that monitor databases to *identify* and *protect* against *database specific attacks* that mostly seek to access sensitive information stored in the databases.

DIAGRAMMATIC REPRESENTATION



SECURITY OF THE DATABASE THROUGH ABSTRACTION

- Data encryption enables to ***encrypt sensitive data***, such as credit card numbers, stored in table columns.
- Encrypted data is ***decrypted*** for a database user ***who has access*** to the data.
- Data encryption helps ***protect data*** stored on ***media*** in the event that the storage media or data file gets stolen.

ADVANTAGES OF DATA ENCRYPTION

- ❑ As a security administrator, one can be sure that sensitive data is safe in case the storage media or data file gets stolen.
- ❑ You do not need to create triggers or views to decrypt data. Data from tables is decrypted for the database user.
- ❑ Database users need not be aware of the fact that the data they are accessing is stored in encrypted form. Data is transparently decrypted for the database users and does not require any action on their part.
- ❑ Applications need not be modified to handle encrypted data. Data encryption/decryption is managed by the database.

AUTHORIZATION

- **Read authorization** - allows reading, but not modification of data
- **Insert authorization** - allows insertion of new data, but not modification of existing data.
- **Update authorization** - allows modification, but not deletion of data.
- **Delete authorization** - allows deletion of data

PRIVILEGES IN DATABASE

select: allows read access to relation, or the ability to query using the view

insert: the ability to insert tuples

update: the ability to update using the SQL update statement

delete: the ability to delete tuples.

PRIVILEGE TO GRANT PRIVILEGES

With grant option:
allows a user who is
granted a privilege to
pass the privilege on
to other users.

Example:

**grant select on
branch to U_1 with
grant option**
gives U_1 the **select**
privileges on *branch* and
allows U_1 to grant this
privilege to others

THANK
YOU